

1. (4 points) Let $\mathbb{Z}^+ = \{x \in \mathbb{Z} \mid x > 0\}$. Define $f : \mathbb{Z}^+ \rightarrow \mathbb{Z}$ by

$$f(z) = \begin{cases} \frac{z}{2}, & \text{if } z \text{ is even,} \\ -\frac{z-1}{2}, & \text{if } z \text{ is odd.} \end{cases}$$

Show that $|\mathbb{Z}^+| = |\mathbb{Z}|$ by checking that f is a bijection.

Injective. Let us assume $f(a) = f(b)$ for $a, b \in \mathbb{Z}^+$. We now should consider cases based on even/odd classifications:

- If a and b are both even, then $f(a) = a/2$ and $f(b) = b/2$, so $a/2 = b/2$ and so we find $a = b$.
- If a and b are both odd, then $f(a) = -(a-1)/2$ and $f(b) = -(b-1)/2$, so

$$-\frac{a-1}{2} = -\frac{b-1}{2} \implies a-1 = b-1 \implies a = b.$$

- If one is even and the other odd, for example a even and b odd, then we find:

$$f(a) = \frac{a}{2} \geq 1 \quad (\text{since } a \geq 2), \quad f(b) = -\frac{b-1}{2} \leq 0.$$

So $f(a) \neq f(b)$. We find this argument still applies to the inverse even/odd assignments of a, b .

Therefore, we find f is injective.

Surjective. Let $y \in \mathbb{Z}$. We wish to show $z \in \mathbb{Z}^+$ with $f(z) = y$.

- If $y \geq 1$, we will let $z = 2y$ - trivially, this is a positive integer. Then, we find:

$$f(z) = f(2y) = \frac{2y}{2} = y.$$

- If $y \leq 0$, we will let $z = 1 - 2y$. Since $y \leq 0$, we have $1 - 2y \geq 1$, and $1 - 2y$ is odd. Then we find

$$f(z) = f(1 - 2y) = -\frac{(1 - 2y) - 1}{2} = -\frac{-2y}{2} = y.$$

Therefore, we find that every $y \in \mathbb{Z}$ has a preimage and so, f is surjective.

Since f is both injective and surjective, it is a bijection, so we find $|\mathbb{Z}^+| = |\mathbb{Z}|$.

2. (8 points) Consider the sequence $(c_n)_{n \geq 0} = (3, 8, 13, 18, 23, \dots)$.

a. **Recursive definition.** Trivially, we find sequence increases by 5 each step, so

$$c_0 = 3, \quad c_{n+1} = c_n + 5 \quad \text{for } n \geq 0.$$

b. **Closed form.** We find this is a sequence with first term 3 and a reoccurring difference of 5, therefore:

$$c_n = 3 + 5n \quad (n \geq 0).$$

c. **Compute c_{52} .** Using the closed form as defined above, we find:

$$c_{52} = 3 + 5 \cdot 52 = 3 + 260 = 263.$$

d. **Is there some n so that $642 = c_n$?** If $642 = c_n$, then $642 = 3 + 5n$, so $5n = 639$ and $n = 639/5 = 127.8$, not an integer. Therefore there is *no* $n \in \mathbb{Z}_{\geq 0}$ such that $642 = c_n$.

3. (4 points) An n -digit number is *good* if it contains the substring “42”, and *bad* otherwise. (No leading zeros.) Let b_n be the number of n -digit bad numbers.

a. **How many bad numbers are there with 2 digits? 3 digits?**

$n = 2$: We find there are 90 two-digit numbers (10 through 99). Trivially, we find the only two-digit number containing “42” is 42 itself. Therefore,

$$b_2 = 90 - 1 = 89.$$

$n = 3$: Similarly, there are 900 three-digit numbers (100 through 999). A three-digit number contains “42” either in positions 1–2 (numbers 420–429, 10 choices) or in positions 2–3 (numbers 142, 242, $\dots$, 942, 9 choices). Since it is impossible for both pairs to equal “42” simultaneously, we find the cases do not overlap and so, the number of good three-digit numbers is $10 + 9 = 19$. Therefore, we have:

$$b_3 = 900 - 19 = 881.$$

b. **Find a recurrence for b_n in terms of b_{n-1} and b_{n-2} (for n large enough).**

We will let x_n be the number of bad n -digit numbers that end in 4, and y_n be the number of bad n -digit numbers that do *not* end in 4. Therefore, we find that $b_n = x_n + y_n$.

We begin by computing x_n . Appending a final digit 4 to any bad $(n-1)$ -digit number cannot create the substring “42” since it would require a 2 *after* the 4, so we find:

$$x_n = b_{n-1}.$$

Now, we wish to compute y_n . To form a bad n -digit number not ending in 4, we will append a digit from $\{0, 1, 2, 3, 5, 6, 7, 8, 9\}$ to a bad $(n-1)$ -digit number, except in cases where the previous number ended in 4, we are not allowed to append 2 - since it would create “42”. Therefore, we have the following equation:

$$y_n = 8x_{n-1} + 9y_{n-1}.$$

We will now use $x_{n-1} = b_{n-2}$ and $y_{n-1} = b_{n-1} - x_{n-1} = b_{n-1} - b_{n-2}$ to produce the following relation:

$$\begin{aligned} b_n &= x_n + y_n \\ &= b_{n-1} + 8x_{n-1} + 9y_{n-1} \\ &= b_{n-1} + 8b_{n-2} + 9(b_{n-1} - b_{n-2}) \\ &= 10b_{n-1} - b_{n-2}. \end{aligned}$$

Therefore, we find a valid recurrence is defined as:

$$b_n = 10b_{n-1} - b_{n-2}$$

where $b_1 = 9$ and $b_2 = 89$, from part (a).

4. (4 points) Recall that for a positive integer n , $n! = n(n-1)(n-2)\cdots 2 \cdot 1$. Use induction to prove that

$$\sum_{i=1}^n i(i!) = (n+1)! - 1 \quad \text{for all positive integers } n.$$

We prove the statement by induction on n .

Base case ($n = 1$):

$$\sum_{i=1}^1 i(i!) = 1 \cdot 1! = 1, \quad (1+1)! - 1 = 2! - 1 = 2 - 1 = 1.$$

Therefore, we find the formula holds for $n = 1$.

Inductive step: Now, we wish to assume that for some $n \geq 1$,

$$\sum_{i=1}^n i(i!) = (n+1)! - 1.$$

Then

$$\begin{aligned} \sum_{i=1}^{n+1} i(i!) &= \left(\sum_{i=1}^n i(i!) \right) + (n+1)((n+1)!) \\ &= ((n+1)! - 1) + (n+1)(n+1)!. \end{aligned}$$

Relatively simply, we find that

$$(n+1)(n+1)! = (n+2-1)(n+1)! = (n+2)! - (n+1)!.$$

Revisiting the previous equality, we find:

$$\begin{aligned} \sum_{i=1}^{n+1} i(i!) &= ((n+1)! - 1) + ((n+2)! - (n+1)!) \\ &= (n+2)! - 1 \\ &= ((n+1)+1)! - 1. \end{aligned}$$

Therefore, we find the statement holds for $n+1$.

Therefore, by induction, we find that $\sum_{i=1}^n i(i!) = (n+1)! - 1$ for all positive integers n .